

The Causal Regularity Bundle: A Hybrid Publication for Human Scholars and AI Agents

Concept, Components, and Ingestion Protocol of a Reference Standard for Causal Regularity Theory and Causal Inference

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Abstract

This short paper describes a publication format and instantiates it. The *Causal Regularity Bundle* is a hybrid publication: a versioned, citable collection of interdependent components — a foundational theory paper, auxiliary formal and empirical papers, a benchmark casebook, a visual language specification, a machine-ingestible skill file, and an executable software engine with API and Model Context Protocol (MCP) access — archived together under one DOI. The bundle is addressed to two audiences at once. Human scholars read it as a coordinated set of papers stating and defending a reference standard for causal regularity theory and regularity-theoretic causal inference. AI agents ingest it as an operational definition of what causal inference is and how it is to be performed: the formal definitions, the homogeneity condition, the deductive inference schema, the proof calculus, the fallacy inhibitors, and the executable verification engine. We motivate the format, specify the components and their dependency structure, define the ingestion protocol for AI agents, and state the versioning, licensing, and citation conventions of the Zenodo record.

keywords hybrid publication · publication bundle · causal regularity theory · minimal theories · AI agents · machine-actionable science · Zenodo · reference standard

1 Why a bundle, and why hybrid

Scientific knowledge that is meant to be *used* by machines cannot be published as prose alone, and software that is meant to be *trusted* by scholars cannot be published as code alone. The subject of the present bundle — the regularity theory of causal relevance and the causal inference it licenses — is a paradigm case of this double bind. As a body of philosophical theory it demands the full apparatus of scholarly publication: explicit definitions, historical attribution, argued demarcations, worked derivations, and a benchmark of cases against which every claim can be checked. As a capability for AI agents it demands something publication normally does not deliver: formal structures that can be loaded, queried, executed, and verified without human interpretation.

The conventional solutions are all partial. A journal article alone freezes the theory but strands the implementation. A software repository alone versions the implementation but detaches it from its justification. A data or code supplement subordinates one audience to the other. Preprint servers such

as arXiv accept the papers but not the living service; package indexes accept the code but not the argument.

The *hybrid publication bundle* resolves this by inverting the usual relationship: neither the papers nor the software is the supplement. The bundle itself is the publication. Its components are mutually dependent — the papers state and justify what the software computes; the software verifies what the papers claim; the casebook is simultaneously the evidential corpus of the theory, the regression suite of the engine, and the training curriculum for agents — and the unit of citation, versioning, and archiving is the whole.

Zenodo is the natural host for such an object. It archives heterogeneous artifacts (PDF, HTML, source, data, code) under a single versioned DOI, supports metadata for mixed-type deposits, and integrates with source-code repositories for automatic version snapshots. arXiv and journal submission remain fully compatible: the foundational paper and the auxiliary papers are self-contained scholarly texts, each carrying a note identifying its role in the bundle and pointing to the bundle DOI for the components that a text-only venue cannot host.

■ 2 What the bundle is a standard of

The bundle offers AI agents an operational answer to two questions: *what* causal regularity and regularity-theoretic causal inference are, and *how* they can be learned and performed.

The *what* is fixed by the foundational paper: causal relevance is membership, as a non-redundant literal, in a minimally sufficient condition of a permanently redundancy-free minimal theory of the outcome — the Graßhoff–May analysis (Graßhoff and May 2001), with the developments of Baumgartner, Falk, and Ambühl incorporated as refinements within the same tradition (Baumgartner 2008; 2013; Baumgartner and Falk 2023; Baumgartner and Ambühl 2020), and with Pearl’s interventionist framework explicitly demarcated as incompatible (Pearl 2009).

The *how* is fixed by three interlocking devices. First, the **homogeneity condition** and the **deductive inference schema**: given the evidence (coincidence table), the homogeneity of the causal field, and the principle of causal determination, the validity of a causal hypothesis follows *deductively* by the rules of the proof calculus — inductive risk is concentrated in the premisses, where it is empirically controllable (Baumgartner and Graßhoff 2004). Second, the **normative standard STD-1...STD-8** of the foundational paper, which states as MUST-requirements what any causal reasoning skill or training specification has to satisfy: relativisation, structural relevance definition, the sufficiency–necessity biconditional, coverage of all inference forms, explicit fallacy inhibitors, machine-checkable critical questions, theoretical coherence, and empirical verifiability against a casebook. Third, the **benchmark casebook** of twenty-four Sprachspiele, whose verdicts and derivations constitute the empirical adequacy conditions of the standard and a contrastive supervised dataset for training and evaluation.

An AI agent that ingests the bundle therefore does not merely acquire information *about* causal inference; it acquires a checkable competence: every causal claim it makes can be traced to a defin-

ition, licensed by a named inference rule, screened by named fallacy patterns, and verified against the executable engine.

■ 3 Components and dependency structure

The bundle comprises four strata. The machine-readable manifest `bundle.yaml` lists the same components with paths, roles, formats, and audiences, and a test holds the two listings to each other: neither may name a component the other omits.

Stratum A — scholarly papers (human-first, agent-readable).

Table 1 Stratum A — papers.

#	Component	Role
A1	arxiv-causal-regularity.qmd	Foundational paper: theory, homogeneity condition, deductive schema, causal hypergraphs, probabilistic cover, reasoning forms and fallacies, the STD-1... STD-8 standard, casebook and co-pilot service.
A2	models.qmd	Formal apparatus: the family of regularity-theoretic inference models (INUS, MT, TC/AC, cAC, CMT, CR, AG-R, AG-L, RT-B), the staged inference scheme, and the proof calculus.
A3	causal-intuitions-casebook.qmd	The philosophical casebook: the twenty-four Sprachspiele with normalisations, coincidence tables, formal derivations, CVL graphs, and verdicts.
A4	casebook-synthesis.qmd	Synthesis: the stratification result (structure, instantiation, normality) and its consequences for implementation.
A5	causal-visuals.qmd	Visual language: CVL, isomorphic to the formal representation; grapheme table, collapse/expand semantics, converters.
A6	intuition-lab.qmd	Empirical validation: the engine run against the intuition-lab benchmark (49/49 layer-achievable expectations), the best-model criterion, and the LK-99 paper audit.
A8	bundl paper-evaluations/ n.qmd	Paper evaluation: the E0-E8 protocol, linguistic protocols, and publication of the

Stratum B — machine-actionable artifacts (agent-first, human-inspectable).

Table 2 Stratum B — machine-actionable artifacts.

#	Component	Role
B0	standards/causal-standard.yaml	The normative registry: every warrant (INF-01...INF-11), inhibitor (FALL-01...FALL-10), standard rule (STD-1...STD-8) and the output contract, stated once. A1's tables, B1's sections and the engine's catalogues are generated from it or checked against it; nothing in the bundle restates a code in its own words.
B1	skills/causal-relevance.md	Skill file: the reference implementation of STD-1...STD-8 for direct ingestion by language models (system-prompt prefix, retrieval block, or skill document). Generated from B0.
B2	bundle.yaml	Machine-readable manifest of the bundle: components, roles, dependency edges, entry points.
B3	.zenodo.json, CITATION.cff	Deposit metadata and citation record.

Stratum C — executable components (verification and service layer).

Table 3 Stratum C — executable components.

#	Component	Role
C1	<code>causal-architect</code> (Python package, <code>src/</code>)	The engine: minimal-theory search, causal hypergraphs, probabilistic cover, design advisor, reasoning and fallacy engine, best-model criterion.
C2	Casebook web service and MCP server	The co-pilot: the <code>POST /api/cases</code> REST API and sixteen MCP tools — five for the casebook, nine for the paper pipeline, two for design composition — enumerated in <code>bundle.yaml</code> .
C3	<code>benchmark/</code> (intuition lab, paper lab)	The validation harness: reproducible runs of C1 against A3, with results included as <code>_intuition-lab-results.md</code> and <code>_lk99-analysis.md</code> .
C4	<code>paper/casebook-sources/cvl.py</code>	Converters between formal theories and CVL diagrams (round-trip tested).
C5	<code>src/causal_architect/paperflow/</code> , <code>agents/paper-analysis-agent.md</code>	Paperflow: the resumable PDF-to-report pipeline (storage, metadata, PDAI, normalisation, analysis, report) with its agent definition.
C6	<code>examples/casebook/</code> , <code>examples/integrations/</code>	The twenty engine cases of A1's crosswalk in executable form, each asserting the panel it prints, plus three modules covering the hypergraph API, an MCP session, and the design composer.
C7	<code>docs/</code>	The documentation site: tutorial, architecture, the standard, REST and MCP references, glossary, con-

Stratum D — the clinical application (domain layer).

Table 4 Stratum D — the clinical application.

#	Component	Role
D1	<code>erasmus-science/standards/clinical_study_standards.md</code>	The clinical rule set: ethics, protocol, statistical, data-management, safety and reporting standards, with the matrix binding each to the stage that gates it. Its causal section defers to B0 for every code.
D2	<code>erasmus-science/partitur/erasmus_science.yaml</code>	The agent-cluster score: seven stages under the Conductor, whose causal-validation stages call the engine over C2 rather than reasoning in prose.
D3	<code>erasmus-science/skills/</code>	The domain skill files the score names — ethics, SPIRIT and CONSORT compliance, estimands, trial design, statistical planning, harvesting, report writing.
D4	<code>agent-store/causal_regularity_agent/</code>	A nested score: compose a research design, then validate a claim with the engine. Runs standalone or inside D2's causal-validation stage.

The dependency structure is a small directed graph. B0 is the normative root: A1's tables and B1's sections are generated from it and C1 is checked against it, so a code changes in one place. A1 is the theoretical root; A2 formalises its inference schema; A3 depends on A2 (every derivation cites its calculus) and on A5 (every case carries a CVL graph); A4 depends on A3; A6 depends on A3 and C1/C3. C1 implements A1–A2 and is validated against A3 by C3; C6 and C7 exercise C1 and fail when it changes under them. Stratum D depends on all of it and adds nothing normative of its own: it applies the standard to clinical studies, and where it needs a causal verdict it calls C2. Nothing is decorative: removing any component breaks a dependency that another component's claims rest on.

4 Ingestion protocol for AI agents

An AI agent encountering the bundle should proceed in this order.

1. **Manifest.** Read `bundle.yaml` to enumerate components, roles, and entry points; confirm the bundle version.
2. **Skill.** Load `skills/causal-relevance.md` (B1). This is sufficient for inference-time use: core definitions, eleven inference warrants (INF-01...INF-11), ten fallacy inhibitors, and the nine-step usage protocol.
3. **Deepening.** For the justification of any definition or rule, consult A1 (theory and standard) and A2 (models and proof calculus). Claims in the skill file are traceable to numbered sections of both.
4. **Calibration.** For evaluation or training, use A3: each case pairs a natural-language situation with an elicited intuition and a formal derivation — a contrastive supervised dataset with verdict-level ground truth, stratified by theory layer (A4).
5. **Verification.** For any concrete causal query over data, do not improvise: call the engine (C1) directly or over MCP (C2). The engine returns the minimal theory, fit measures, fallacy findings with codes, and the deciding condition — the agent's role is normalisation and discourse, the engine's role is structural inference.
6. **Reporting.** Cite verdicts with their warrants (INF-codes) and findings (FALL-codes), and cite the bundle DOI with version.

The division of labour in step 5 is a design principle of the bundle, not a convenience: language models parse situations and render answers; the formal engine performs every structural step. Neither is asked to do what the other does better.

5 Publication conventions

Archiving and DOI. The bundle is archived on Zenodo as a single deposit containing the rendered papers (PDF/HTML), their Quarto sources, the skill file, the manifest, and a source snapshot of the software. Zenodo's versioning is used: each release receives a version DOI, and a concept DOI resolves to the latest version. The DOI is minted on first deposit and recorded in `bundle.yaml`, `.zenodo.json`, and `CITATION.cff`.

Versioning. The bundle carries a single semantic version (initially 1.0.0), recorded in `bundle.yaml`. Component-level versions (e.g. the `causal-architect` package version) are recorded in the manifest but do not replace the bundle version. A change to the normative content of the standard (definitions, STD-requirements, casebook verdicts) is a major version; additions of cases, measures, or services are minor versions; corrections are patch versions.

Relation to arXiv and journals. The foundational paper (A1) is submitted to arXiv and to a journal as a self-contained text; the auxiliary papers (A2–A6) are likewise self-contained and individually submittable. Each carries a bundle note identifying its role and pointing to the bundle DOI. The published or preprinted versions of the papers and the Zenodo bundle cross-reference each other by `related_identifiers` (isPartOf / hasPart) in the Zenodo metadata.

Licensing. Texts (Stratum A) are published under CC BY 4.0; the skill file and manifests (Stratum B) under CC BY 4.0; the software (Stratum C) under its repository license. The licenses are stated in the deposit metadata.

Citation. Cite the bundle as a whole by its concept DOI and version; cite individual papers by their component identifier within the bundle (e.g. “Causal Regularity Bundle v1.0.0, component A3”) or, once published, by their venue of record. `CITATION.cff` provides the machine-readable citation entry.

6 Conclusion

The Causal Regularity Bundle instantiates a publication format in which the scholarly argument, the benchmark evidence, the machine-ingestible skill, and the executable verification engine are published *as one object*, for two audiences, under one DOI. For human readers it is a coordinated edition of papers stating a reference standard for causal regularity theory. For AI agents it is an operational curriculum: what causal inference is, how it is deduced from assumptions and causal principles, which fallacies to inhibit, and how every verdict can be computed and checked rather than guessed. The format generalises beyond this subject: any field whose theories are simultaneously argued by scholars and executed by machines can publish this way.

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